FS10

Firmware Architecture and Service Guide

USB-to-FS10 bridge behavior, monitoring logic, and service reference

Firmware source	firmware/FS10-ATMEGA32U4/FS10-ATMEGA32U4.ino
Controller	ATmega32U4
Purpose	Explain operating modes, data flow, local monitoring, and service boundaries

System role - The FS10 monitor sits between a PC and the FS10 instrument. It forwards the PC serial session while independently presenting local status on the TFT display.

1. Interfaces and operating boundary

The firmware uses the ATmega32U4 native USB CDC interface for the PC connection and Serial1 at 9600 baud, 8N1 for the FS10 connection through the MAX232E RS-232 converter. The bridge forwards PC and FS10 traffic without modifying its payload.

- USB CDC interface: PC-facing transport at 115200 baud.
- Serial1 interface: FS10-side serial transport at 9600 baud, 8N1.
- Display interface: SPI TFT display for local operating state, readouts, graph, serial number, and firmware build identifier.
- Dedicated inputs: USB presence, 24 V presence, and the two relay/contact status signals.

2. Mode selection

The firmware selects its operating behavior from the USB-presence input.

Condition	Firmware behavior
USB absent	Stand-alone monitor mode: reads FS10 data, performs scheduled configuration polling, requests measurements, and updates the local display.
USB present	USB bridge mode: forwards PC-to-FS10 and FS10-to-PC traffic. Local private polling is deferred until the bridge is quiet.

Transition behavior - A USB or supply-state change resets mode-specific polling state and redraws the monitor screen. The display header changes between STAND ALONE MONITOR and USB BRIDGE PASSTHRU.

3. Stand-alone monitor behavior

When USB is absent, the firmware reads incoming FS10 data and performs background polling only when the serial bridge is quiet. A configuration pass requests CUST_FLOW_MIN, CUST_FLOW_MAX, RELAY_LIMIT, reference temperature, dR, and the FS10 serial number; periodic measurement requests use the *meas command.

Returned serial lines are parsed into flow, temperature, switch point, dR, and serial-number display values. Flow history is converted to a 0 to 100 percent scale using the stored configuration limits and plotted on the TFT graph.

4. USB bridge behavior

With USB present, the bridge handles traffic in both directions in bounded 32-byte chunks. PC bytes are forwarded to Serial1. FS10 bytes are parsed for local display data and forwarded to the USB CDC interface unless they are part of a firmware-private refresh response.

- Private polling waits for a quiet bridge interval before requesting the FS10 serial number or dR value.
- A serial number refresh is queued after a USB attach or a recovered cable condition.
- USB write failures, PC-to-FS10 traffic, and FS10-to-PC traffic drive distinct RGB LED activity indications.
- The TFT update is deferred while serial traffic is active, reducing display work during bridge activity.

Bridge priority - The PC serial session is the primary transport while USB is present. Local polling occurs only in an idle interval.

5. Relay, supply, and cable-status logic

Relay state is evaluated independently of the serial stream. The firmware reads PF0 and PF1 directly and recognizes two valid states: PF0 high / PF1 low is LO; PF1 high / PF0 low is HI. The other two combinations are treated as unknown.

- An unknown relay state must persist for the cable-fault debounce interval before the cable warning is shown.
- Loss of the 24 V supply clears readings and presents the supply warning screen.
- Relay HI/LO transitions drive a two-tone beep sequence and update the TFT status region.
- The firmware polls these inputs from the main loop; it does not use GPIO interrupts for relay state.

6. Local display and service information

The TFT presents Flow, Temp, Switch Pt., dR, a trend graph, HI/LO status, and the FS10 serial number. The firmware build identifier is shown at the bottom of the screen in the form FW build YYYY-MM-DD HH:MM.

ICSP1 supports AVRISP mkII programming. The Arduino build and upload procedure is documented separately in the FS10 Programming Guide. PCB layout, power-path, USB-protection, copper-pour, and hardware-interface details are documented separately in the FS10 PCB Hardware Design Description.

7. Document boundary

This guide describes firmware behavior and service-relevant interfaces. The firmware source, KiCad schematic, KiCad PCB, Programming Guide, and PCB Hardware Design Description remain the controlling technical references for their respective subjects.